Lesson Overview: Data Engineering in Spark

In today's data-centric environment, mastering data manipulation and optimisation with Apache Spark is crucial when handling large-scale data. This is vital for addressing quality issues, ensuring data security, and preparing data for analytical tasks. In this lesson, we'll delve into using Apache Spark for fundamental and advanced data engineering tasks, enhancing our capability to manage and transform data efficiently.



What You'll Learn:

▼**Understand and interpret dataset structures**

Extract and parse data to identify and understand the structure and content of datasets.

▼Transform data for operational and analytical use

Translate and map data to conform to specific operational and analytical needs.

▼Filter, aggregate and summarize datasets

Conduct filtering, aggregation, and summarisation to distill essential information and insights from data.

▼Improve data quality through enhancement

Enhance datasets through enrichment and imputation to address data quality issues.

▼Optimize retrieval with indexing and ordering

Utilise indexing and ordering to improve data organization and retrieval efficiency.

▼Apply privacy-preserving techniques

Implement anonymisation and encryption to ensure data privacy and meet regulatory requirements.

▼Standardize and model data

Apply data modeling, typecasting, formatting, and renaming to enhance data usability.

▼Reshape data for analysis

Pivot data to transform its layout for better analysis and reporting capabilities.

Ready to Dive In?

Reflect:
What data engineering tasks have you previously completed using PySpark or SQL? How might Spark's distributed computing framework enhance your current workflows?